

Método bilineal: $s = \frac{z}{T} \frac{z-1}{z+1} = \frac{z}{T} \frac{1-z^{-1}}{1+z^{-1}}$

$C(s) = k_p + k_i/s + k_d \cdot s \rightarrow$ general

$C(z) = \frac{u(z)}{e(z)} = k_p + k_i \frac{T}{z} \cdot \frac{1+z^{-1}}{1-z^{-1}} + k_d \cdot \frac{z}{T} \frac{1-z^{-1}}{1+z^{-1}}$

Proporcional: $C(z) = k_p = \frac{u(z)}{e(z)} \Rightarrow u(m) = k_p \cdot e(m)$

PI: $k_d = 0 \rightarrow C(z) = k_p + k_i \frac{T}{z} \cdot \frac{1+z^{-1}}{1-z^{-1}} = \frac{u(z)}{e(z)}$

Multiplico por $(1-z^{-1})$ y por $e(z)$

$u(z) \cdot (1-z^{-1}) = \left[k_p (1-z^{-1}) + k_i \frac{T}{z} (1+z^{-1}) \right] \cdot e(z)$

$$u(z)(1 - z^{-1}) = \left[\left(k_p + k_i \frac{T}{2} \right) + \left(-k_p + k_i \frac{T}{2} \right) z^{-1} \right] e(z)$$

Antitransforme:

$$u(n) - u(n-1) = \underbrace{\left(k_p + k_i \frac{T}{2} \right)}_{a_0} e(n) + \underbrace{\left(-k_p + k_i \frac{T}{2} \right)}_{a_1} e(n-1)$$

$$u(n) = [a_0 \quad a_1] \begin{bmatrix} e(n) \\ e(n-1) \end{bmatrix} + u(n-1)$$

$$\text{PD)} \quad c(s) = k_p + k_d \cdot s$$

$$c(z) = k_p + k_d \frac{z}{T} \frac{1 - z^{-1}}{1 + z^{-1}} = \frac{u(z)}{e(z)}$$

$$\rightarrow u(z) \cdot (1 + z^{-1}) = e(z) \cdot \left[k_p + k_d \cdot \frac{z}{T} (1 - z^{-1}) \right]$$

$$\Rightarrow u(z)(1 + z^{-1}) = e(z) \left[k_p + k_d \frac{z}{T} - k_d \frac{z}{T} z^{-1} \right]$$

Antitransforma:

$$u(m) + u(m-1) = \left(k_p + \frac{2}{T} k_d \right) e(m) - \frac{2}{T} k_d e(m-1)$$

$$u(m) = [a_0 \ a_1] \begin{bmatrix} e(m) \\ e(m-1) \end{bmatrix} - u(m-1)$$

Puede ser más fácil hacerlo como lo hace Pose:

$$P: u(m) = k_p e(m)$$

integral anterior

$$I: u(m) = \frac{T}{2} \cdot e(m) + \frac{T}{2} e(m-1) + u(m-1) = \frac{T}{2} e(m) - \frac{T}{2} e(m-1) + I(m-1)$$

$$\Rightarrow u(m) = \frac{T}{2} e(m) + \frac{T}{2} e(m-1) + I(m-1)$$

$$D: u(m) = \frac{2}{T} (e(m) - e(m-1)) - D(m-1)$$

$$i.e. PI: u(m) = k_p \cdot e(m) + k_i \cdot I(m), \text{ con } I(m) = \frac{T}{2} e(m) + \frac{T}{2} e(m-1) + I(m-1)$$

$$i.e. PD: u(m) = k_p e(m) + k_d \cdot D(m), \text{ con } D(m) = \frac{2}{T} e(m) - \frac{2}{T} e(m-1) - D(m-1)$$